

This is a **Physical Delivery Hub** that uses ADA guidelines to create signage and a shelving unit as a foundation. By integrating tactile markers and solar lighting, we create a predictable, accessible drop-off point. This project bridges the gap between digital navigation and the built environment, ensuring that every student can retrieve their food delivery orders independently and with dignity.

Verification of Use: We verified this solution by testing the unit's 30" vertical height. By lifting orders off the ground, we eliminate the need for students to bend down (a critical accessibility barrier) and provide a consistent, sanitized target for drivers. The steel wire construction was stress-tested to ensure it remains stable under the weight of multiple heavy delivery bags.

Product & Tool Overview

- **Shelve Unit Price:** \$29.99
 - **Dimensions For shelf:** 23.3" W × 13.4" D × 30" H
 - **Required Tools: None.** This kit is designed for tool-free assembly.
 - **Recommended Tools - Rubber Mallet:** To seat shelves firmly without damaging the black finish
 - **High Visibility Reflective Tape Price:** \$5.99
 - **Length For Tape:** 1inx30ft
 - **Acrylic Board:** \$15
 - **Dimensions For Acrylic Board:** 24inx18in
 - **Zip Ties:** \$20
 - **CO2 Laser Cutter**
 - Minimum 40W power, 24" x 12" bed size
 - **Vinyl Plotter**
 - Minimum 12" cutting width
 - **Squeegee**
 - For bubble-free vinyl application
 - **Digital Calipers**
 - To verify material thickness for hole tolerances
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Assembly Instructions

1. Level the Foundation

Screw the **leveling feet** into the bottom of the four poles. Since campus entryways or sidewalks can be uneven, these feet allow you to stabilize the hub. Start with them screwed all the way in and adjust later if the unit wobbles

2. Set the Height Markers

- The poles have horizontal grooves spaced at 1-inch intervals
- Snap two halves of the **plastic slip sleeves** together around each pole at your desired height
- **Tactile Feedback:** Slide the sleeve until you feel the interior rib "click" into a groove
- **Orientation:** Ensure the "TOP" arrow on the sleeve points up. All four sleeves must be at the same height for a level shelf

3. Install the First Tier

Slide the four poles through the corner cylinders of the first wire shelf

- Lower the shelf onto the sleeves. The cylinders will wedge the sleeves against the pole, locking them in place
- **Pro Tip:** Push down firmly on each corner to ensure the shelf is fully seated

4. Final Tiers & Accessibility

Repeat the process for the middle and top shelves.

- **Strategic Placement:** We recommend placing the middle shelf at a height reachable for wheelchair users. The top shelf (30" high) serves as the primary drop-off point to keep food at a comfortable "hand-off" height

Design Files

- **Sign_Base_Main.STEP:** The master 3D file for CNC/3D printing
- **Laser_Cut_Profile.DXF:** Optimized for 2D laser cutting; includes 4mm zip-tie slots.
- **Braille_Grid_Template.SVG:** A 1:1 scale map for dot placement

Detailed Step-by-Step Sign

Path 1: Laser-Only (The "Quick-Build")

- **Material Prep:** If using plywood, sand the surface with 220-grit paper before cutting
- **Laser Settings:** * *Cut:* Speed 10mm/s, Power 90% (based on 60W laser)
 - *Engrave (Optional):* Speed 300mm/s
 - Power 20%
- **The Cut:** Load the 23-inch file. Ensure the zip-tie holes (slotted for adjustability) are cut clean
 - If using acrylic, flame-polish the edges with a torch for a premium look.

Path 2: Laser + Vinyl (High-Contrast Multi-Color)

- **Base:** Laser cut the 23" base out of white or clear acrylic
- **Vinyl Preparation:** Cut your text in a "negative" style
 - the letters are holes in the vinyl
- **Color Taping:** On the *face* of the acrylic base, lay down strips of colored electrical or duct tape where the letters will sit.
- Clean the base with 70% Isopropyl Alcohol
 - Apply the vinyl overlay using transfer tape
 - The tape colors will "pop" through the letter-shaped holes in the vinyl
- **Trimming:** Trim any excess tape from the edges of the sign with a utility knife

Path 3: Braille & Tactile (ADA-Inspired)

- **Substrate Cut:** Cut the standard 23" base
 - **Braille Mapping:**
 - *Option A (Machine):* Program the laser to "score" (low power) small crosshairs where dots belong
 - *Option B (Manual):* Print the SVG template at 100% scale and tape it to the board as a drill/glue guide
 - **Dot Application:** Use a toothpick to apply a tiny drop of super glue to each crosshair
 - Place the 3mm tactile dots
 - *Pro Tip:* Use a "wax picker" tool to handle the tiny dots easily
 - **The Contrast Coat:** Spray the entire assembly with matte paint. The physical height of the dots remains tactile (at least 0.6mm high), while the uniform color ensures the sign doesn't look "cluttered"
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Mounting & Maintenance

- **Alignment:**
 - Hold the sign against the shelf edge
 - Feed a zip tie through the top-left hole, loop it around the shelf wire, and close it loosely
 - Repeat for the other three corners
 - Use a spirit level on the top edge of the sign before pulling the zip ties tight
 - Snip the excess "tails" of the zip ties flush to the sign to avoid sharp edges
 - Wipe with a damp microfiber cloth
 - Avoid ammonia-based cleaners on acrylic to prevent "crazing" (fine cracks)
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Hub Implementation & Safety

- **ADA Integration:** Once the frame is built, attach your ADA-compliant signage and high-contrast tactile markers to the shelf edges.
- **Visibility:** Mount solar-powered lighting to the top poles to guide drivers during night-time deliveries.
- **Ballast:** For outdoor placement at Brandeis, place a heavy object (like a sandbag or brick) on the **bottom shelf** to lower the center of gravity and prevent tipping in high winds.
- **Sanitation:** The black powder coating is easy to clean; wipe the wire tiers regularly with a damp cloth to maintain a food-safe environment.